

Experiment analysis step 3: period and order diagnostics

Session-period checks for the primary crossover sample

W241 Plank Study

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0.1 Purpose

This document checks whether broad period effects or order patterns appear large enough to threaten interpretation of the main crossover estimate.

These diagnostics are descriptive first. They do not replace the primary model; they help explain whether session order or repeated exposure appears material and whether the period term in the fixed-effects model is doing useful work.

0.2 Load analytic datasets

```
paired <- sanitize_dt_ascii(fread(derived_path("paired_analysis.csv")))
long_dt <- sanitize_dt_ascii(fread(derived_path("long_repeated_measures.csv")))

long_primary <- long_dt[in_primary_analysis == 1L]

stopifnot(uniqueN(long_primary$participant_id) == nrow(paired))
stopifnot(nrow(long_primary) == 2L * nrow(paired))
```

0.3 Period summaries

```
period_summary <- long_primary[
  ,
  .(
    n = .N,
    mean_tte = mean(outcome_tte),
    sd_tte = sd(outcome_tte),
    median_tte = median(outcome_tte),
    iqr_tte = IQR(outcome_tte)
```

```

),
  by = period
][order(period)]

period_summary[, `:=`(
  mean_tte = round(mean_tte, 3),
  sd_tte = round(sd_tte, 3),
  median_tte = round(median_tte, 3),
  iqr_tte = round(iqr_tte, 3)
)]

kable(period_summary, caption = "Session 1 vs Session 2 summaries")

```

Table 1: Session 1 vs Session 2 summaries

period	n	mean_tte	sd_tte	median_tte	iqr_tte
1	38	93.334	46.730	84.646	68.303
2	38	97.178	55.207	70.692	75.268

0.4 Period-by-order summaries

```

order_period_summary <- long_primary[
  ,
  .(
    n = .N,
    mean_tte = mean(outcome_tte),
    sd_tte = sd(outcome_tte),
    mean_rpe = mean(post_rpe, na.rm = TRUE)
  ),
  by = .(crossover_order, period, treatment)
][order(crossover_order, period)]

order_period_summary[, `:=`(
  mean_tte = round(mean_tte, 3),
  sd_tte = round(sd_tte, 3),
  mean_rpe = round(mean_rpe, 3)
)]

kable(order_period_summary, caption = "Period-by-order diagnostic summary")

```

Table 2: Period-by-order diagnostic summary

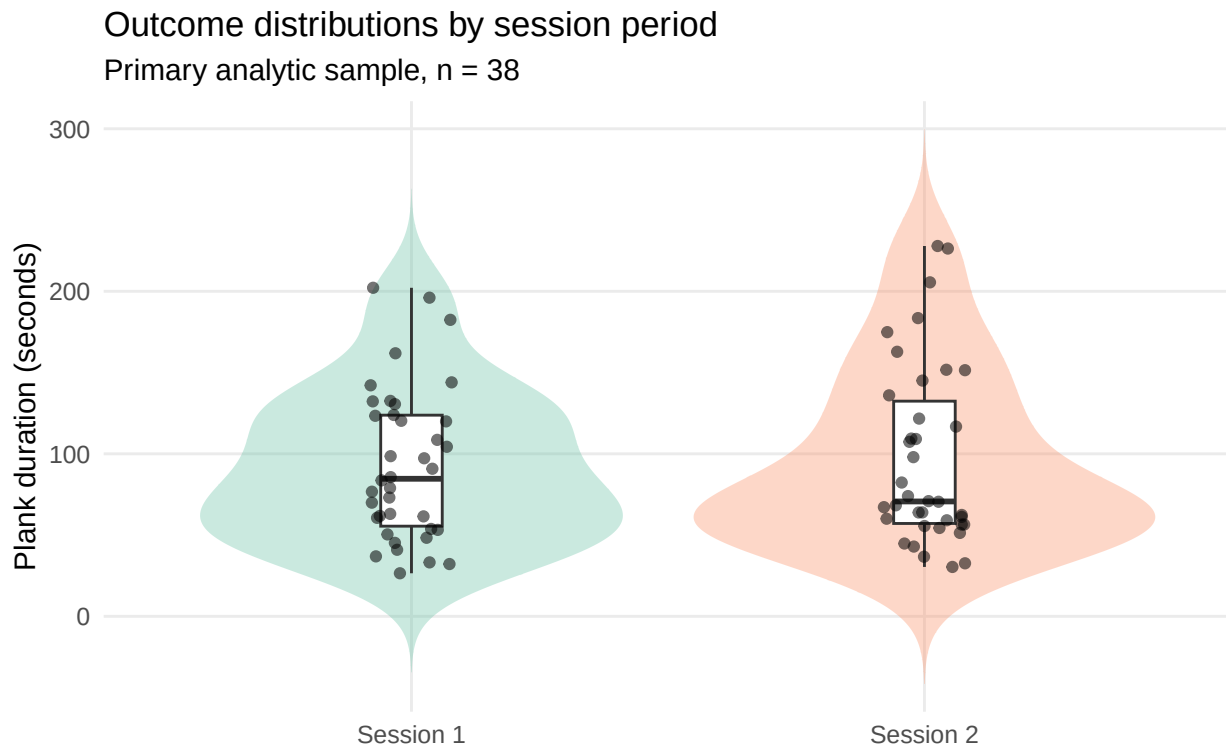
crossover_order	period	treatment	n	mean_tte	sd_tte	mean_rpe
high_tempo_first	1	high_tempo	19	91.001	44.336	6.895
high_tempo_first	2	low_tempo	19	80.476	46.395	6.263
low_tempo_first	1	low_tempo	19	95.666	50.116	7.105
low_tempo_first	2	high_tempo	19	113.880	59.359	7.632

0.5 Session-period figures

0.5.1 Session 1 vs Session 2 boxplot

```
long_primary[, period_label := factor(period, levels = c(1, 2), labels = c("Session 1", "Session 2"))]

ggplot(long_primary, aes(x = period_label, y = outcome_tte, fill = period_label)) +
  geom_violin(alpha = 0.35, color = NA, trim = FALSE) +
  geom_boxplot(width = 0.12, fill = "white", outlier.shape = NA) +
  geom_jitter(width = 0.08, height = 0, alpha = 0.55, size = 1.6) +
  scale_fill_brewer(palette = "Set2", guide = "none") +
  labs(
    title = "Outcome distributions by session period",
    subtitle = paste0("Primary analytic sample, n = ", uniqueN(long_primary$participant_id)),
    x = NULL,
    y = "Plank duration (seconds)"
  ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid.minor = element_blank())
```



0.5.2 Order-arm trajectories across periods

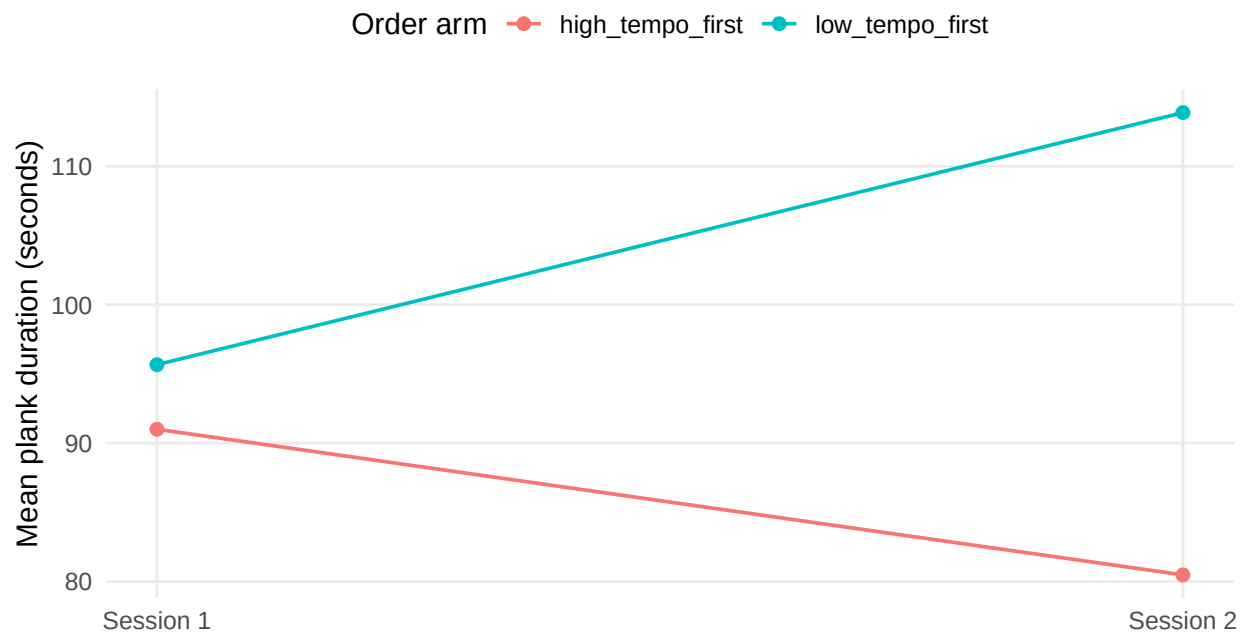
```
order_plot <- long_primary[
  ,
  .(mean_tte = mean(outcome_tte)),
  by = .(crossover_order, period)
]

ggplot(order_plot, aes(x = period, y = mean_tte, color = crossover_order, group = crossover_order)) +
```

```
geom_line(linewidth = 0.7) +
geom_point(size = 2.1) +
scale_x_continuous(breaks = c(1, 2), labels = c("Session 1", "Session 2")) +
labs(
  title = "Average plank time by order arm across periods",
  subtitle = "Checks for sequence or learning patterns",
  x = NULL,
  y = "Mean plank duration (seconds)",
  color = "Order arm"
) +
theme_minimal(base_size = 12) +
theme(panel.grid.minor = element_blank(), legend.position = "top")
```

Average plank time by order arm across periods

Checks for sequence or learning patterns



0.6 Diagnostic models

0.6.1 Broad period check

```
period_check <- t.test(
  long_primary[period == 2, outcome_tte],
  long_primary[period == 1, outcome_tte],
  paired = FALSE
)

period_check_tab <- data.table(
  quantity = c(
    "Mean Session 2",
    "Mean Session 1",
    "Difference (Session 2 - Session 1)",
    "95% CI lower",
  )
)
```

```

    "95% CI upper",
    "p-value"
  ),
  value = c(
    mean(long_primary[period == 2, outcome_tte]),
    mean(long_primary[period == 1, outcome_tte]),
    diff(rev(period_check$estimate)),
    period_check$conf.int[1],
    period_check$conf.int[2],
    period_check$p.value
  )
)

period_check_tab[, value := round(value, 4)]
kable(period_check_tab, caption = "Broad period comparison")

```

Table 3: Broad period comparison

quantity	value
Mean Session 2	97.1777
Mean Session 1	93.3336
Difference (Session 2 - Session 1)	3.8441
95% CI lower	-19.5456
95% CI upper	27.2339
p-value	0.7441

0.6.2 Sequence diagnostic regression

```

long_primary[, crossover_order := factor(crossover_order)]
long_primary[, period_factor := factor(period)]

diag_fit <- lm(
  outcome_tte ~ period_factor * crossover_order,
  data = long_primary
)

diag_vcov <- sandwich::vcovCL(diag_fit, cluster = ~ participant_id, type = "HC1")
diag_ct <- lmtest::coeftest(diag_fit, vcov. = diag_vcov)

diag_terms <- intersect(
  c("period_factor2", "crossover_orderlow_tempo_first", "period_factor2:crossover_orderlow_tempo_first"),
  rownames(diag_ct)
)

diag_results <- data.table(
  term = diag_terms,
  estimate = diag_ct[diag_terms, "Estimate"],
  std_error = diag_ct[diag_terms, "Std. Error"],
  p_value = diag_ct[diag_terms, "Pr(>|t|)"]
)

diag_results[, `:=`(
  estimate = round(estimate, 4),

```

```

    std_error = round(std_error, 4),
    p_value = round(p_value, 4)
  )]

kable(diag_results, caption = "Sequence diagnostic regression")

```

Table 4: Sequence diagnostic regression

term	estimate	std_error	p_value
period_factor2	-10.5253	5.8264	0.0750
crossover_orderlow_tempo_first	4.6650	15.4541	0.7636
period_factor2:crossover_orderlow_tempo_first	28.7389	9.4895	0.0034

0.7 Interpretation notes

```

period_effect <- round(diag_ct["period_factor2", "Estimate"], 3)
interaction_effect <- round(diag_ct["period_factor2:crossover_orderlow_tempo_first", "Estimate"], 3)

message("Estimated broad period effect (Session 2 vs Session 1): ", period_effect, " seconds")
## Estimated broad period effect (Session 2 vs Session 1): -10.525 seconds
message("Estimated period-by-order interaction: ", interaction_effect, " seconds")
## Estimated period-by-order interaction: 28.739 seconds

```

0.8 Verification

```

stopifnot(uniqueN(long_primary$participant_id) == 38L)
stopifnot(all(long_primary$period %in% c(1L, 2L)))
stopifnot(nrow(order_period_summary) == 4L)

message("Diagnostic sample participants: ", uniqueN(long_primary$participant_id))
## Diagnostic sample participants: 38
message("Diagnostic sample rows: ", nrow(long_primary))
## Diagnostic sample rows: 76

```